Lecture 11

Scaling – case study and details

CS336

Motivation today

What is the best practices for scaling and hparam tuning LMs?

- Does chinchilla’s approach to scaling actually work?
- Can we save compute when training and fitting these things?
- Should we be picking particular architectures / parametrizations to scale nicely?

Scaling in practice

The newest model we talked about with scaling details - 2022

DeepMind

Training Compute-Optimal Large Language Models

Jordan Hoffmann*, Sebastian Borgeaud*, Arthur Mensch*, Elena Buchatskaya, Trevor Cai, Eliza Rutherford, Diego de Las Casas, Lisa Anne Hendricks, Johannes Welbl, Aidan Clark, Tom Hennigan, Eric Noland, Katie Millican, George van den Driessche, Bogdan Damoc, Aurelia Guy, Simon Osindero, Karen Simonyan, Erich Elsen, Jack W. Rae, Oriol Vinyals and Laurent Sifre*

*Equal contributions

What about more recently?

deepseek

DeepSeek LLM
Scaling Open-Source Language Models with Longtermism

2024

MiniCPM: Unveiling the Potential of Small Language Models with Scalable Training Strategies

Shengding Hu[1], Yuge Tu[2], Xu Han[1]*, Chaoqun He[1], Ganqu Cui[1], Xiang Long[2], Zhi Zheng[2], Yewei Fang[2], Yuxiang Huang[1], Weilin Zhao[1], Xinrong Zhang[1], Zheng Leng Thai[1], Kaihuo Zhang[2], Chongyi Wang[2], Yuan Yao[1], Chenyang Zhao[1], Jie Zhou[2], Jie Cai[2], , Zhongwu Zhai[2], Ning Ding[1], Chao Jia[2], Guoyang Zeng[2], Dahai Li[2], Zhiyuan Liu[1]*, Maosong Sun[1]
[1]Department of Computer Science and Technology, Tsinghua University.
[2]Modelbest Inc.
shengdinghu@gmail.com

2024

Meta

The Llama 3 Herd of Models

Llama Team, AI @ Meta[1]
[1]A detailed contributor list can be found in the appendix of this paper.

2024

Hunyuan-Large: An Open-Source MoE Model with 52 Billion Activated Parameters by Tencent

Tencent Hunyuan Team

2024

MINIMAX

MiniMax-01: Scaling Foundation Models with Lightning Attention

MiniMax[1]

2025

KIMI K2: OPEN AGENTIC INTELLIGENCE

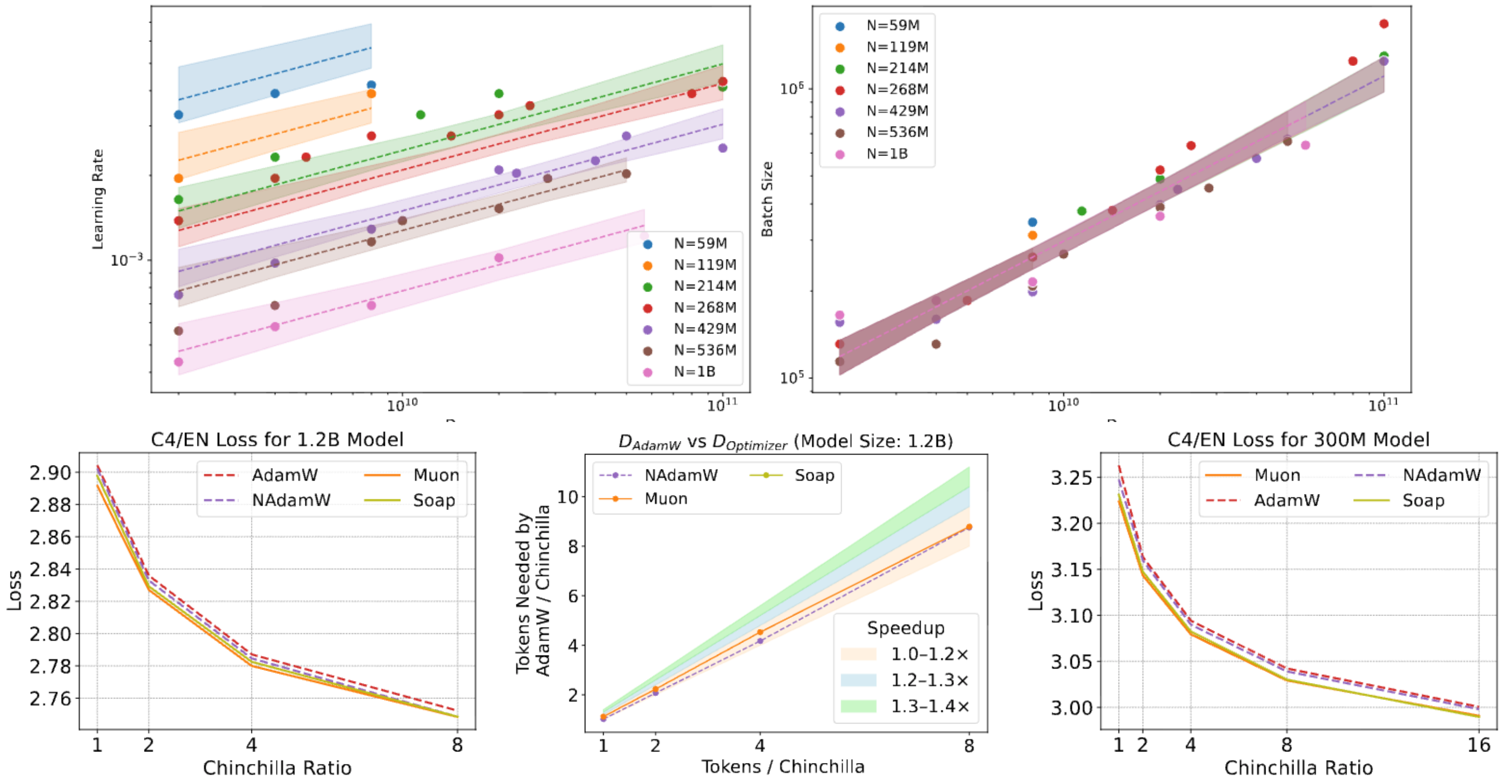
TECHNICAL REPORT OF KIMI K2

Kimi Team

2026

Initialization and optimizers

Initialization, optimizers, and various hyperparams (LR/batch) can be scale sensitive



Working through some detailed, public scaling recipes

1. MiniCPM

MiniCPM: Unveiling the Potential of Small Language Models with Scalable Training Strategies

Shengding Hu[1], Yuge Tu[2], Xu Han[1]*, Chaoqun He[1], Ganqu Cui[1], Xiang Long[2], Zhi Zheng[2], Yewei Fang[2], Yuxiang Huang[1], Weilin Zhao[1], Xinrong Zhang[1], Zheng Leng Thai[1], Kaihuo Zhang[2], Chongyi Wang[2], Yuan Yao[1], Chenyang Zhao[1], Jie Zhou[2], Jie Cai[2], , Zhongwu Zhai[2], Ning Ding[1], Chao Jia[2], Guoyang Zeng[2], Dahai Li[2], Zhiyuan Liu[1]*, Maosong Sun[1]
[1]Department of Computer Science and Technology, Tsinghua University.
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2. DeepSeek

deepseek

DeepSeek LLM
Scaling Open-Source Language Models with Longtermism

MiniCPM

MiniCPM (2024) – small, high-perf LM from Tsinghua group.

MiniCPM: Unveiling the Potential of Small Language Models with Scalable Training Strategies

Shengding Hu[1], Yuge Tu[2], Xu Han[1]*, Chaoqun He[1], Ganqu Cui[1], Xiang Long[2], Zhi Zheng[2], Yewei Fang[2], Yuxiang Huang[1], Weilin Zhao[1], Xinrong Zhang[1], Zheng Leng Thai[1],Kaihuo Zhang[2], Chongyi Wang[2], Yuan Yao[1], Chenyang Zhao[1], Jie Zhou[2], Jie Cai[2], , Zhongwu Zhai[2], Ning Ding[1], Chao Jia[2], Guoyang Zeng[2], Dahai Li[2], Zhiyuan Liu[1]*, Maosong Sun[1]

[1]Department of Computer Science and Technology, Tsinghua University.
[2]Modelbest Inc.
shengdinghu@gmail.com

Careful, extensive scaling computations + muP to stabilize and simplify scaling

Not really 'sota' model (even in 2024) but many interesting lessons on scaling

MiniCPM

High performance 1-2.5 B parameter models. These models beat most out 2Bs and match many modern 7B models.

Model	C-Eval	CMMLU	MMLU	HumanEval	MBPP	GSM8K	MATH
Llama2-7B	32.42	31.11	44.32	12.20	27.17	13.57	1.80
Qwen-7B	58.96	60.35	57.65	17.07	42.15	41.24	5.34
Deepseek-7B	42.82	44.45	47.82	20.12	41.45	15.85	1.53
Mistral-7B	46.12	42.96	62.69	27.44	45.20	33.13	5.00
Gemma-7B	42.57	44.20	60.83	38.41	50.12	47.31	6.18
Llama2-13B	37.32	37.06	54.71	17.07	32.55	21.15	2.25
MPT-30B	29.34	32.09	46.56	21.95	35.36	10.31	1.56
Falcon-40B	40.29	41.57	53.53	24.39	36.53	22.44	1.92
TinyLlama-1.1B	25.02	24.03	24.3	6.71	19.91	2.27	0.74
Qwen-1.8B	49.81	45.32	43.37	7.93	17.8	19.26	2.42
Qwen1.5-1.8B	**55.00**	50.85	43.81	5.49	24.82	26.16	3.25
Gemini Nano-3B	-	-	-	-	27.20	22.80	-
StableLM-Zephyr-3B	30.34	30.89	45.90	35.37	31.85	52.54	12.12
Phi-2(2B)	23.37	24.18	52.66	47.56	**55.04**	**57.16**	3.50
Gemma-2B	29.26	28.56	38.49	24.39	29.74	16.83	3.34
MiniCPM-1.2B	49.14	46.81	49.63	44.51	32.75	31.77	10.60
MiniCPM-2.4B	51.13	**51.07**	**53.46**	**50.00**	47.31	53.83	**10.24**

Techique 1: muP to stabilize scaling

Scale_emb = 12, scale_depth = 1.4, init_std = 0.1, lr =0.01

Name	Specific Operation
Embedding Output Scaling	Multiply the output of the embedding by *scale_emb*
Residual Connection Scaling	Scale the increment at each residual connection in each layer by $scale\_depth/\sqrt{\text{num\_layers}}$
Initialization of Tensors	Set the initialization standard deviation of each two-dimensional tensor parameter to $init\_std/\sqrt{d_m/d_{base}}$, and set other parameters' initialization to 0.1
Learning Rate Scaling of Tensors	Adjust the learning rate of each two-dimensional tensor parameter to $1/(d_m/d_{base})$ times the learning rate of other parts (or the overall learning rate)
LM Head Scaling	Adjust the output logits to $1/(d_m/d_{base})$ times the original value

Table 7: List of operations used when applying tensor program techniques.

Scaling recipe / strategy

Use muP for initialization, fix the aspect ratio, scale up the overall model size.

Name	**N (B)**	d_m	d_{ff}	d_h	n_h	L
9M	0.009	320	800	64	5	8
30M	0.036	512	1280	64	8	12
70M	0.066	640	1600	64	10	14
0.1B	0.109	768	1920	64	12	16
0.17B	0.166	896	2240	64	14	18
0.2B	0.241	1024	2560	64	16	20
0.5B	0.499	1344	3360	64	21	24

Note that the gap between the largest model here and the actual model they train is ~5x

Optimal batch, LR, token-to-size ratios are directly fitted via scaling analysis

Optimal LR

According to muP – optimal learning rate should be (roughly) stable. Is it?

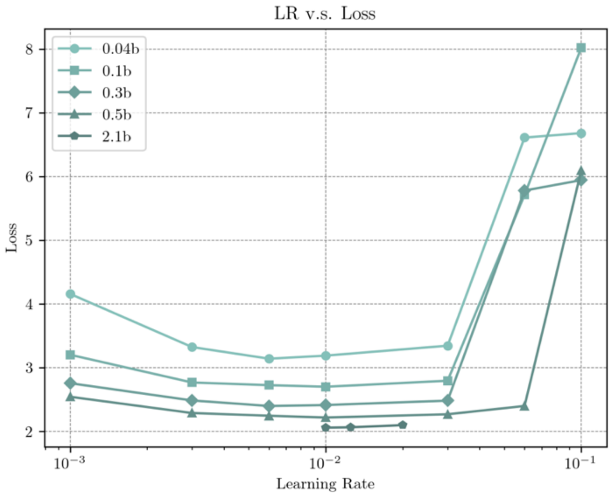
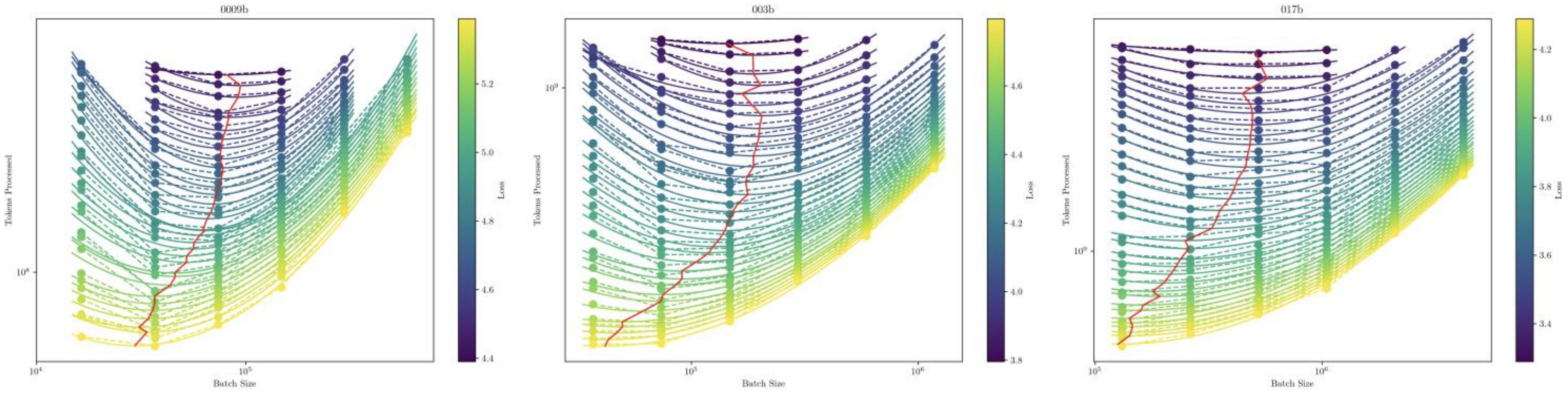


Figure 3: Loss vs Learning Rate. After applying for the Tensor Program, the learning rate shift becomes minimal.

Optimal batch

Three model sizes (9m, 30m, 170m) as a function of data size (y), batch (x) and loss (col)

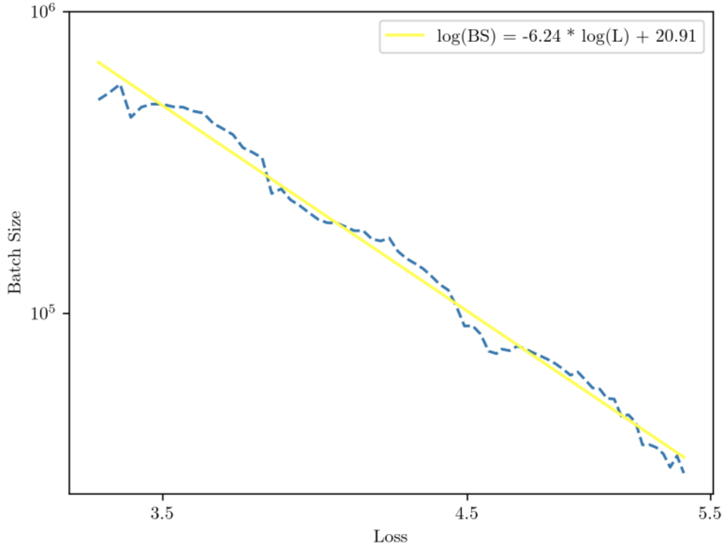


Vertical columns of points represent a single training curve (fixed batch, more points).

Red line connects the minima of parabolic fits to equal-loss points.
For a fixed model and target loss, it selects the batch size requiring the fewest training tokens.

Optimal batch size

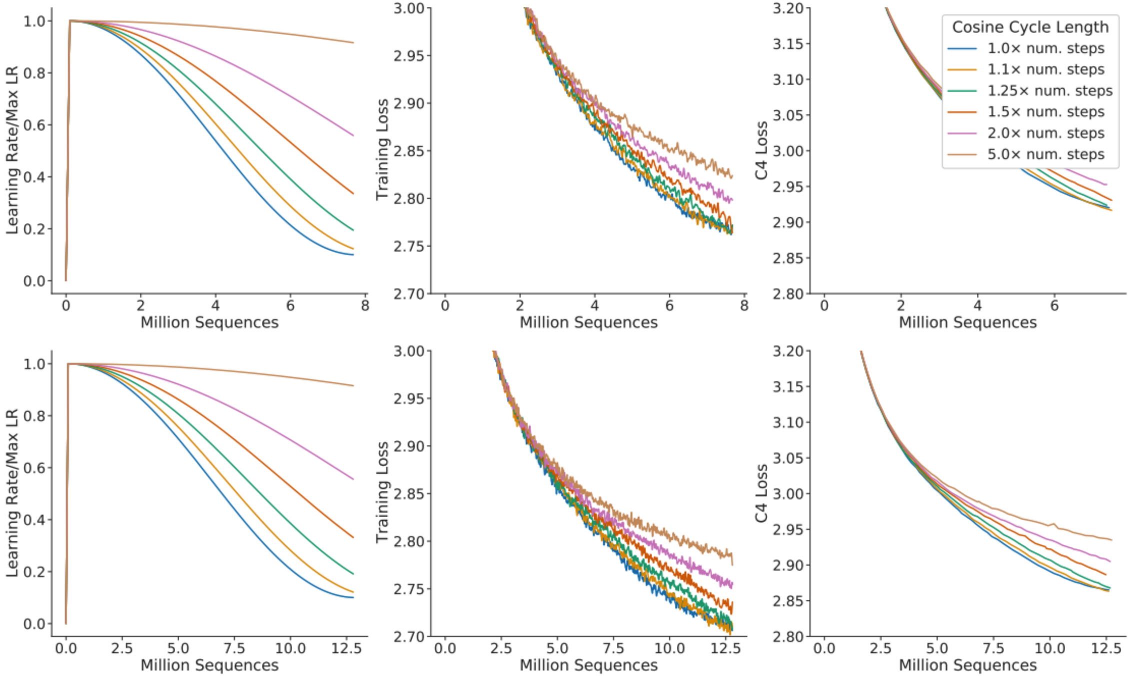
We can then follow the Kaplan 2020 analysis and plot optimal batch size vs final loss.



Fairly clean trend – polynomially increase the batch size as loss decreases.

What remains – model size vs data tradeoffs.

From chinchilla – to fit a scaling law, we need to train from scratch, not just early stop



This turns the cost of fitting a scaling law from n to n^2.. Can we avoid this?

(partial) solution in miniCPM – WSD learning rate

Instead of cosine, split learning rate into warmup, stable, and decay phases.

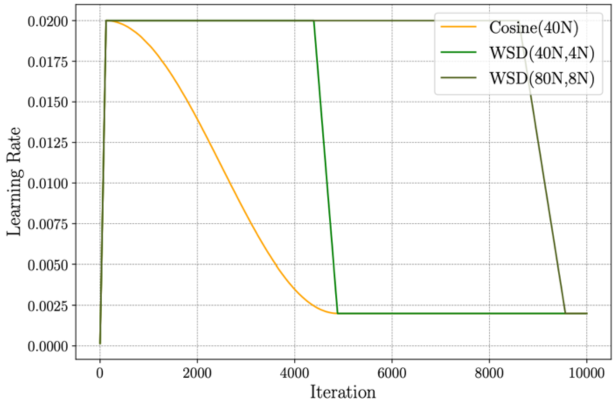


Figure 15: Illustrative comparison between Cosine LRS and WSD LRS. The WSD LRS with different end steps share the same stable training stage.

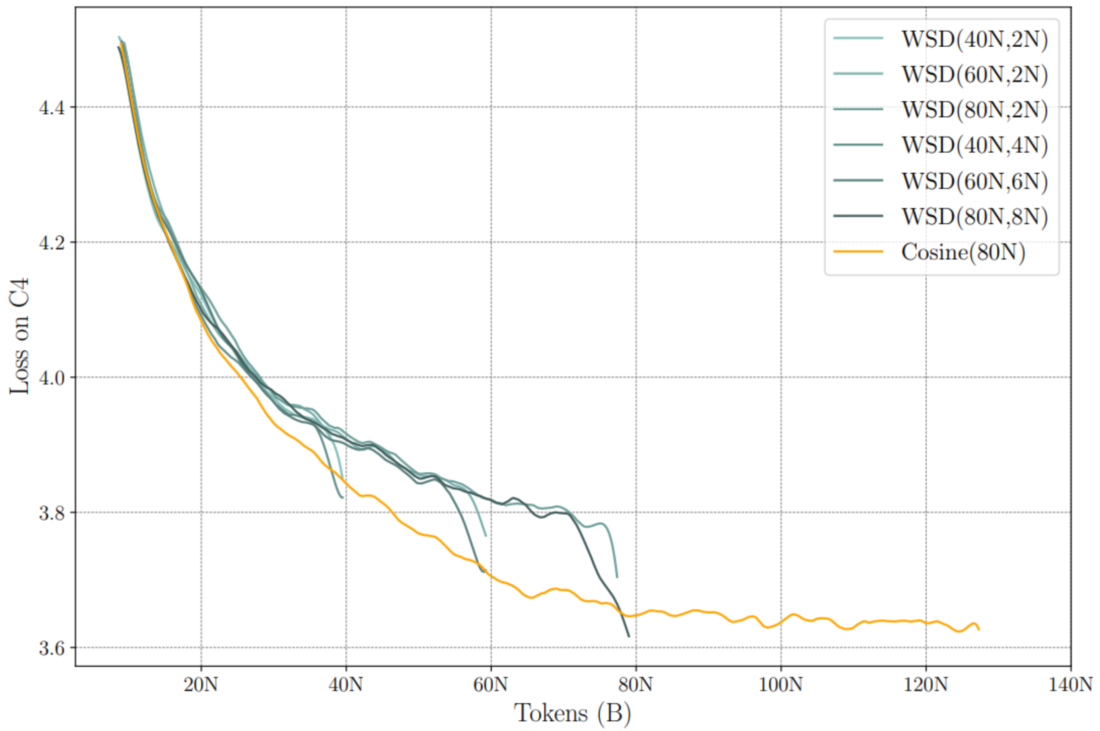
In this section, we introduce the utilization of the WSD scheduler as an effective approach to explore the scaling law with linear cost ($O(mC)$). Since WSD scheduler has the advantage of arriving at the optimal loss of Cosine LRS after decaying from stable stage's checkpoints of any step, we are now able to precisely measure the optimal scaling properties without re-training the models from scratch to different amount of tokens, thus making the scaling law measurement much more efficient along the data axis.

We measure the scaling law along the data and model axes by training SLMs of 6 sizes ranging from 0.04B to 2B, each with 6 decayed model starting from checkpoint of $10N$ to $60N$ data during the stable training stage. The final loss is evaluated on five held-out evaluation dataset. To potentially compare the loss when the model uses different tokenizer, we take the average of loss by number of bytes instead of number of tokens, following Achiam et al. (2023). The final loss of each pair of data size and model size is shown in the blue lines in Figure 17.

For chinchilla-style analysis, can restart the run at the end of the stable phase.

WSD learning rates work well in miniCPM

Slower during the stable phase, rapid loss decay during decay phase. Decay ~ 10%.



Chinchilla-type analysis

Equipped with the WSD learning rate,
we can now try to find the optimal data-to-model size ratio

Then we fit the losses with model size N and data size D following Hoffmann et al. (2022) using scipy `curvefit` function:

$$L(N, D) = C_N N^{-\alpha} + C_D D^{-\beta} + L_0 \tag{2}$$

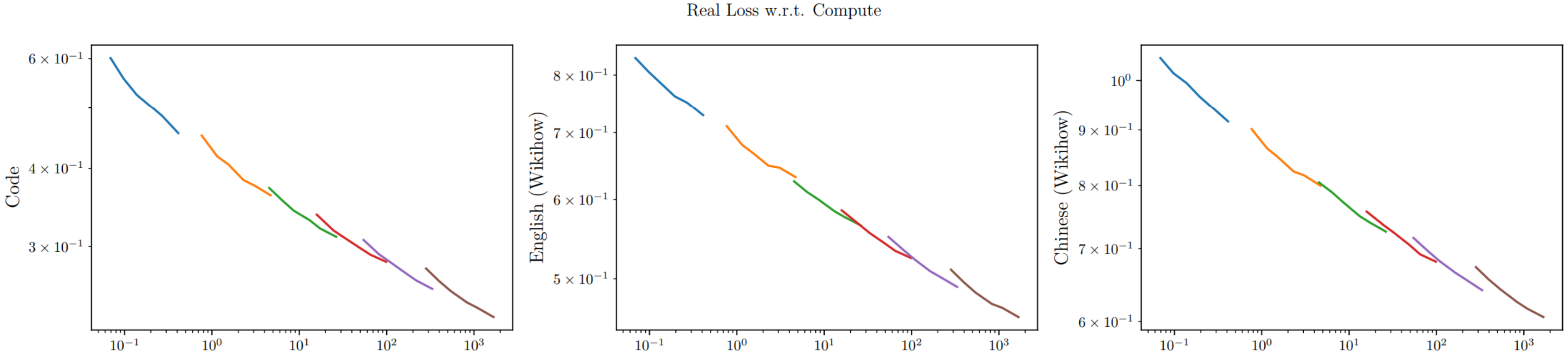
The fitted curve along the data axis for each dataset and each checkpoints are shown in orange lines in Figure 17. Then we have the optimal model size N_{opt}, dataset size D_{opt}, given a fixed amount of compute $C = 6ND$ (Rae et al., 2021) as:

$$\frac{N_{opt}}{D_{opt}} = K^2 \left(\frac{C}{6}\right)^{\eta}, \tag{3}$$

MiniCPM authors choose method 1 (lower envelope) and method 3 (joint fit)

Chinchlla method 1

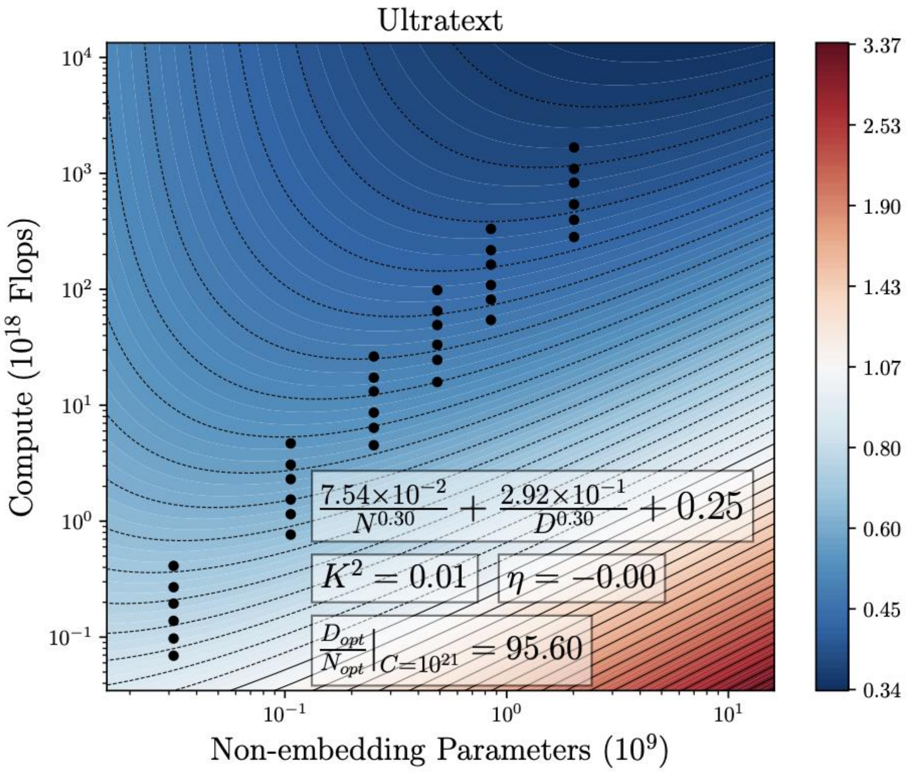
Fairly clear (though maybe not linear?) trends



Different colors indicate different models. Their runs suggest relatively low diminishing returns due to data.

Chinchilla method 3

Their primary scaling approach is the joint fit – they find *very* high data-model ratios.



DeepSeek

DeepSeek (2024) – another LM with careful scaling analysis

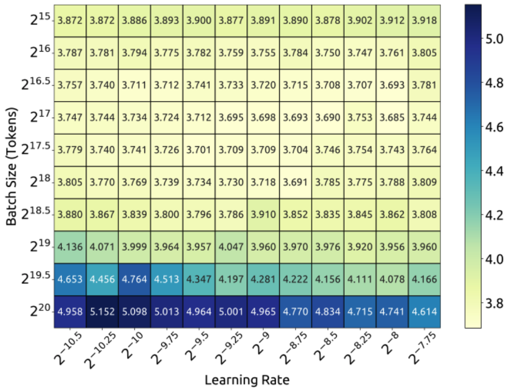
DeepSeek LLM
Scaling Open-Source Language Models with Longtermism

7 and 67B param models – generally high performance compared to other open LM

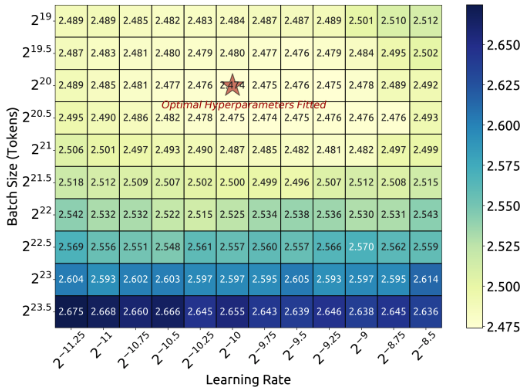
Scaling strategy – batch + LR

Scaling strategy: don’t use any muP, directly estimate optimal batch / LR

We initially conducted a grid search for batch size and learning rate on small-scale experiments with a compute budget of 1e17, and the results of a specific model size (177M FLOPs/token) are illustrated in Figure 2(a). The results demonstrate that the generalization error remains stable across a wide range of choices of batch sizes and learning rates. This indicates that near-optimal performance can be achieved within a relatively wide parameter space.



(a) 1e17 FLOPs (177M FLOPs/token)

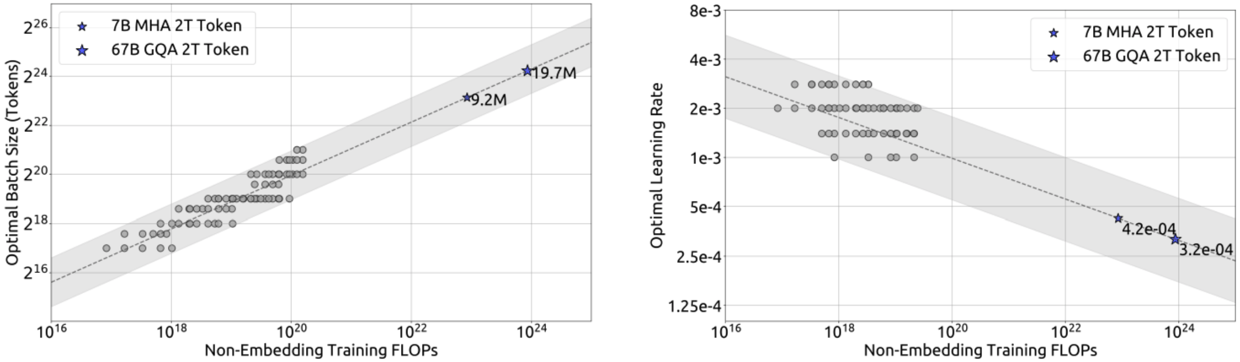


(b) 1e20 FLOPs (2.94B FLOPs/token)

Scaling analysis of learning rates

Small scale runs + collect ‘near optimal’ (within 0.25% of min) models.

$$\eta_{\text{opt}} = 0.3118 \cdot C^{-0.1250}$$
$$B_{\text{opt}} = 0.2920 \cdot C^{0.3271} \tag{1}$$



(a) Batch size scaling curve

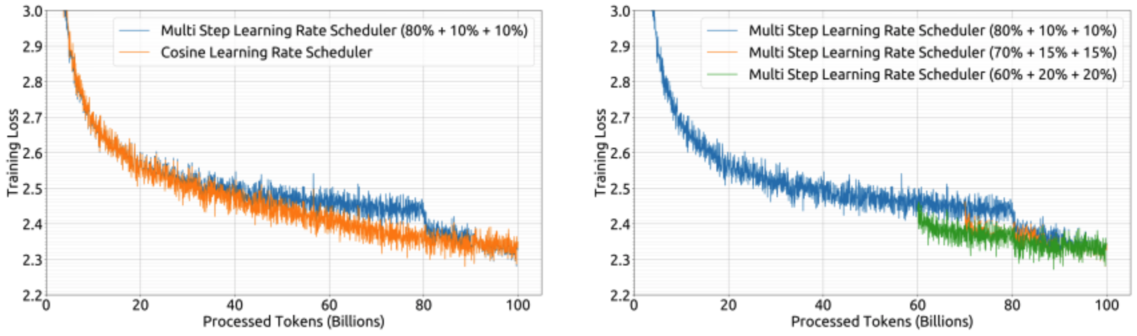
(b) Learning rate scaling curve

Learning rate fit looks a bit questionable..

For chinchilla analysis: WSD-style learning rate

Deepseek uses WSD-style learning rate – fast warmup + two decay steps of 10% each.

A multi-step learning rate scheduler is employed during pre-training instead of the typical cosine scheduler. Specifically, the learning rate of the model reaches its maximum value after 2000 warmup steps, and then decreases to 31.6% of the maximum value after processing 80% of the training tokens. It further reduces to 10% of the maximum value after 90% of the tokens. The gradient clipping during the training phase is set to 1.0.



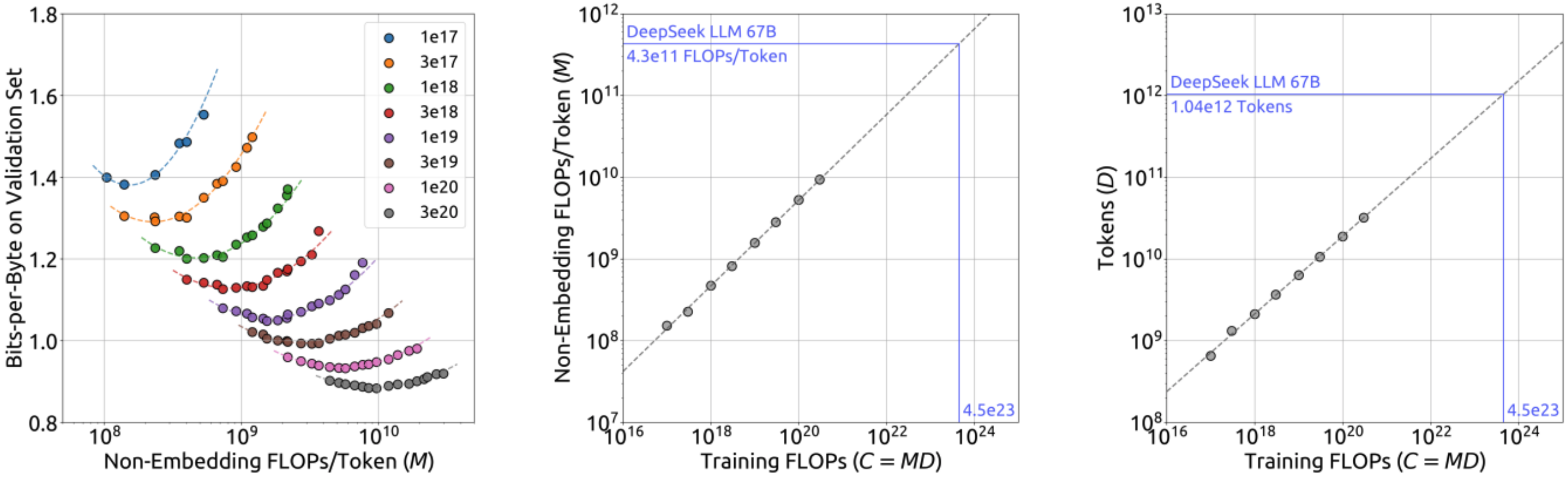
(a) Multi-step v.s. cosine learning rate decay

(b) Different proportions of multi-step stages

Generally seems to match performance of cosine learning rates.

Data-size tradeoff analysis: Chinchilla method 2

Straightforward isoflop-style analysis for selecting the model size tradeoffs.



(a) IsoFLOP curve (b) Optimal model scaling (c) Optimal data scaling

Scaling predicts final model loss

The fitted scaling models (generally) accurately predict the final model losses.

Additionally, we fitted the loss scaling curve according to compute budget C and optimal generalization error, and predicted the generalization error for DeepSeek LLM 7B and 67B, as shown in Figure 5. The results indicate that using small-scale experiments can accurately predict

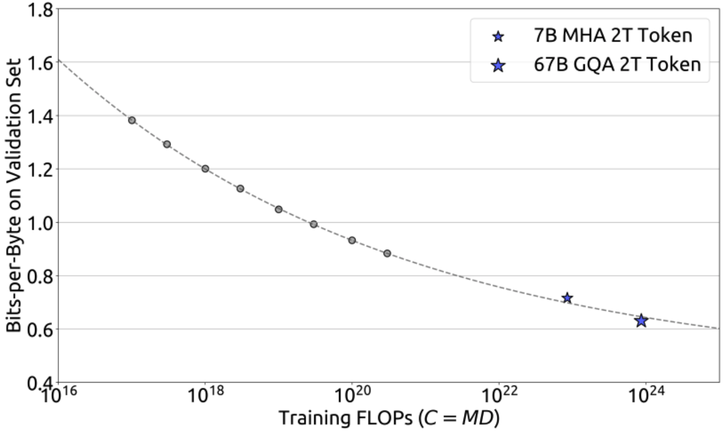


Figure 5 | Performance scaling curve. The metric is the bits-per-byte on the validation set. The dotted line represents the power law fitting the smaller model (grey circles). The blue stars represent DeepSeek LLM 7B and 67B. Their performance is well-predicted by the scaling curve.

Other models: Qwen scaling

Qwen 2.5 – batch and hyper scaling fits

3.2 Scaling Law for Hyper-parameters

We develop scaling laws for hyper-parameter based on the pre-training data of Qwen2.5 (Hoffmann et al., 2022; Kaplan et al., 2020). While previous studies (Dubey et al., 2024; Almazrouei et al., 2023; Hoffmann et al., 2022) primarily used scaling laws to determine optimal model sizes given compute budgets, we leverage them to identify optimal hyperparameters across model architectures. Specifically, our scaling laws help determine key training parameters like batch size B and learning rate μ for both dense models and MoE models of varying sizes.

Through extensive experimentation, we systematically study the relationship between model architecture and optimal training hyper-parameters. Specifically, we analyze how the optimal learning rate μ_{opt} and batch size B_{opt} vary with model size N and pre-training data size D. Our experiments cover a comprehensive range of architectures, including dense models with 44M to 14B parameters and MoE models with 44M to 1B activated parameters, trained on datasets ranging from 0.8B to 600B tokens. Using these optimal hyper-parameter predictions, we then model the final loss as a function of model architecture and training data scale.

Additionally, we leverage scaling laws to predict and compare the performance of MoE models with varying parameter counts against their dense counterparts. This analysis guides our hyper-parameter configuration for MoE models, enabling us to achieve performance parity with specific dense model variants (such as Qwen2.5-72B and Qwen2.5-14B) through careful tuning of both activated and total parameters.

Qwen 3 – similar scaling for LR/batch

Similar to Qwen2.5 (Yang et al., 2024b), we develop scaling laws for optimal hyper-parameters (e.g., learning rate scheduler, and batch size) predictions based on three pre-training stages mentioned above. Through extensive experiments, we systematically study the relationship between model architecture, training data, training stage, and optimal training hyper-parameters. Finally, we set the predicted optimal learning rate and batch size strategy for each dense or MoE model.

Kimi K2

Many recent papers – sparsity scaling law to find the right sparsity levels

Sparsity Scaling Law We develop a sparsity scaling law tailored for the Mixture-of-Experts (MoE) model family using Muon. Sparsity is defined as the ratio of the total number of experts to the number of activated experts. Through carefully controlled small-scale experiments, we observe that — under a fixed number of activated parameters (i.e., constant FLOPs) — increasing the total number of experts (i.e., increasing sparsity) consistently lowers both the training and validation loss, thereby enhancing overall model performance (Figure 5). Concretely, under the compute-optimal sparsity scaling law, achieving the same validation loss of 1.5, sparsity 48 reduces FLOPs by 1.69×, 1.39×, and 1.15× compared to sparsity levels 8, 16, and 32, respectively. Though increasing sparsity leads to better performance, this gain comes with increased infrastructure complexity. To balance model performance with cost, we adopt a sparsity of 48 for Kimi K2, activating 8 out of 384 experts per forward pass.

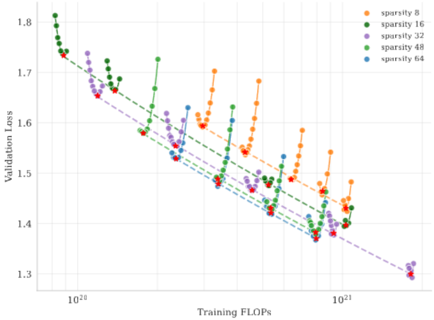


Figure 5: Sparsity Scaling Law. Increasing sparsity leads to improved model performance. We fixed the number of activated experts to 8 and the number of shared experts to 1, and varied the total number of experts, resulting in models with different sparsity levels.

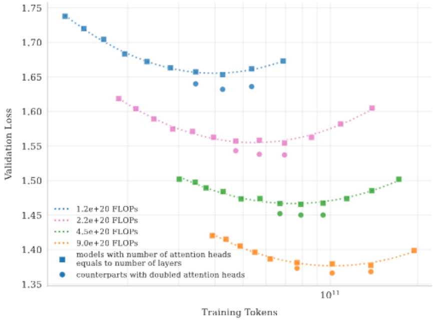


Figure 6: Scaling curves for models with number of attention heads equals to number of layers and their counterparts with doubled attention heads. Doubling the number of attention heads leads to a reduction in validation loss of approximately 0.5% to 1.2%.

Hunyuan (2024) large scaling laws

Yet more isoflops-style scaling (but this time for MoE parameter sizes)

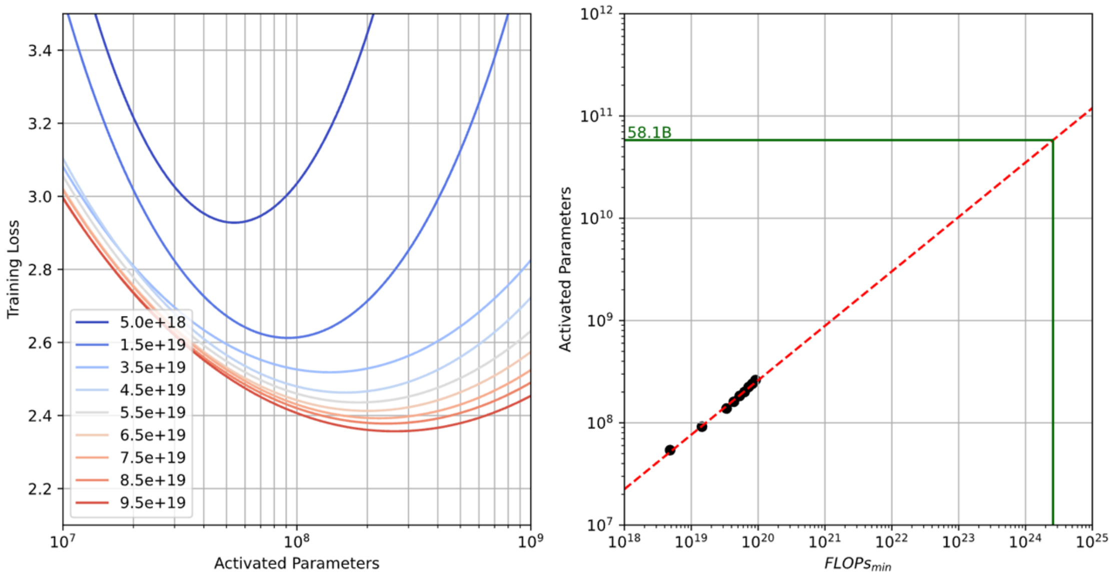


Figure 3: Using quadratic polynomial fitting, we obtain the scaling law of the optimal number of activation parameters under different minimum compute budgets.

Optimal ratio – 96-1 (data to active param)

LLaMA 3 (2024) Scaling laws

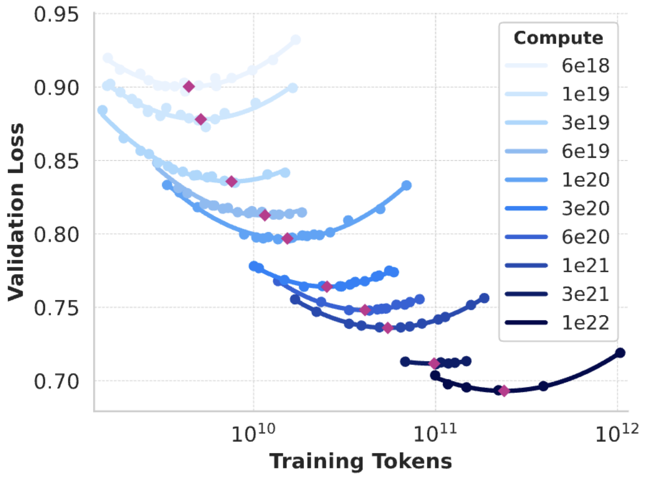
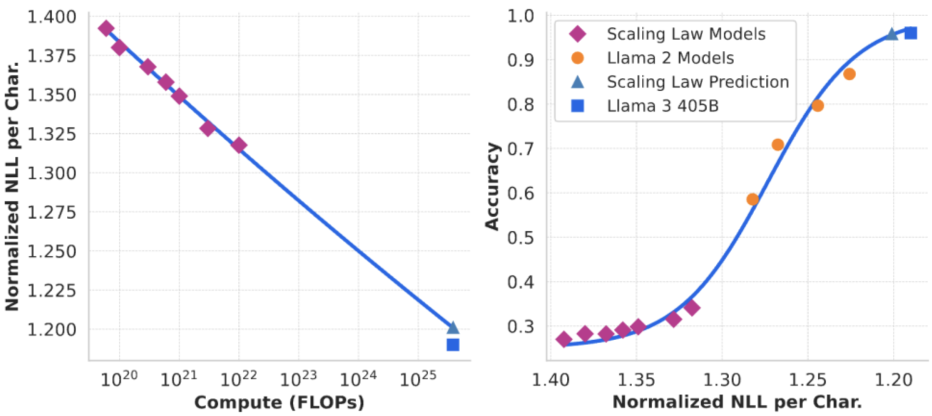


Figure 2 Scaling law IsoFLOPs curves between 6×10^{18} and 10^{22} FLOPs. The loss is the negative log-likelihood on a held-out validation set. We approximate measurements at each compute scale using a second degree polynomial.



Isoflops-style scaling (39-1 ratio)

Compute-to-downstream scaling

MiniMax-01 (2025)

Architecture scaling laws + Chinchilla method 1

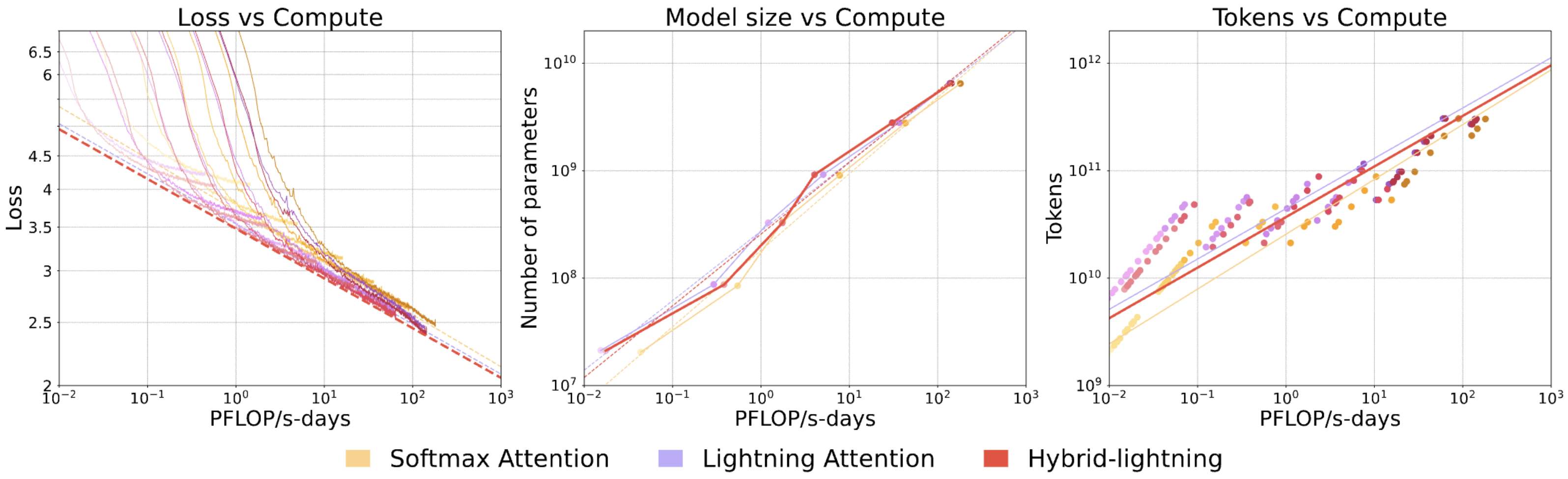


Figure 6 | **Summary of Scaling Laws.** Training curves (left) span models from 70M to 7B parameters. Optimal model size (center) and training tokens (right) are derived based on a specified compute budget estimation.

Recent scaling law recipes

DeepSeek recipe

- Assume most transformer hypers are invariant to scale
- Do a scaling analysis on batch / LR to figure out optimal scaling
- IsoFLOP analysis to figure out model sizing
  - Use a piecewise-linear schedule to make chinchilla scaling cheap.

miniCPM recipe

- Use muP to make transformer + LR invariant to scale
- Use a piecewise linear schedule to get sample for Chinchilla method 3 (curve fitting)

Recent (late 2024+) but less detailed

Qwen – Lr/batch (very few details..)

Kimi K2 – MoE scaling

LLaMA 3 / Hunyuan

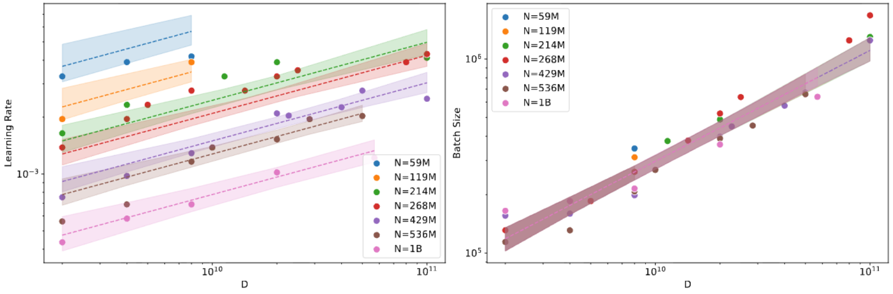
- Just isoflops (no other scaling details)

Minimax

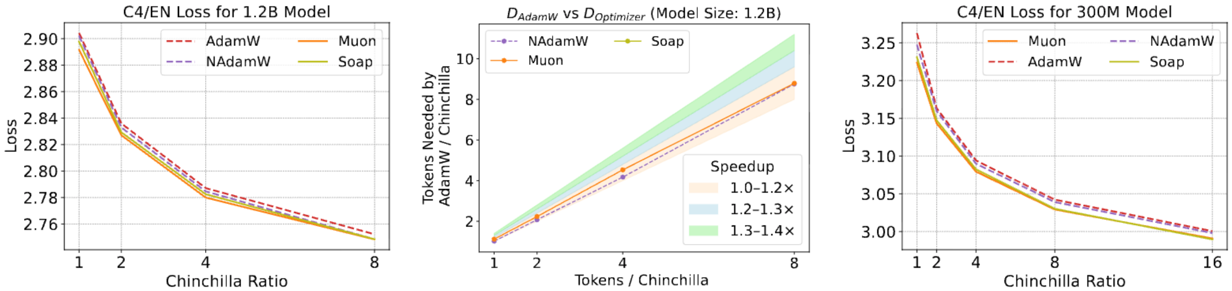
- Architecture choice / decision scaling

Optimizer scaling

Optimizers choices / tuning can be tricky and scale sensitive



How should we pick different optimizers?



StepFun scaling law study

Large scale scaling law study from StepFun

Predictable Scale: Part I, Step Law – Optimal Hyperparameter Scaling Law in Large Language Model Pre-training

Houyi Li* StepFun, Fudan University
Wenzhen Zheng* StepFun
Qiufeng Wang StepFun
Hanshan Zhang StepFun
Zili Wang StepFun
Shijie Xuyang StepFun, Fudan University
YuanTao Fan StepFun
Zhenyu Ding Xi'an Jiaotong University
Haoying Wang Xi'an Jiaotong University
Ning Ding Xi'an Jiaotong University
Shuigeng Zhou Fudan University
Xiangyu Zhang StepFun, Megvii Technology
Daxin Jiang StepFun

Core question: how do we set LR / batch params as we scale? (Deepseek / qwen approach)

Core question – scaling rates on LR and batch

What are the right variables / functional form?

Name	Data Recipe	Model Sparsity	Learning Rate	Batch Size	Relative Error
OpenAI Law [20]	✘	✘	$3.239 * 10^{-3} + -1.395 * 10^{-4} log(N)$	$2e18\mathcal{L}^{-4.76190}$	9.51‰
Microsoft Law [2]	✘	✘	$1.3192e^{-5} N^{-0.23} D^{-0.32}$	-	9.25‰
DeepSeek Law [6]	✘	✘	$0.3188C^{-0.1250}$	$0.2920C^{0.3271}$	9.26‰
Porian Law [26]	✘	✘	$3.7N^{-0.36}$	$0.7576N^{0.703}$	3.71‰
MiniCPM Law [18]	✘	✘	-	$\frac{2e18}{L^{6.24}}$	-
MeiTuan Law [36]	✘	✔	$\lambda\mathcal{L}^{-\alpha}$	$\lambda_B\mathcal{L}^{-\alpha_B^{-1}}$	-
Ours (Step Law)	✔	✔	$1.79N^{-0.713}D^{0.307}$	$0.58D^{0.571}$	**0.94‰**

Different views:

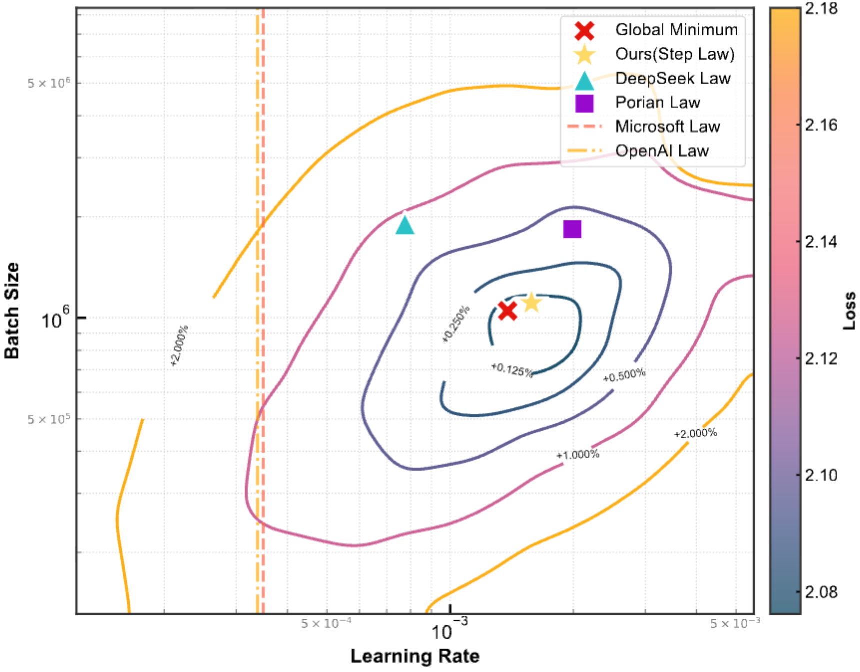
Critical batch: batch as a function of loss (OpenAI)

Compute power law: poly function of compute (DeepSeek)

.. Or something else?

The approach: purely empirical – grid search the space

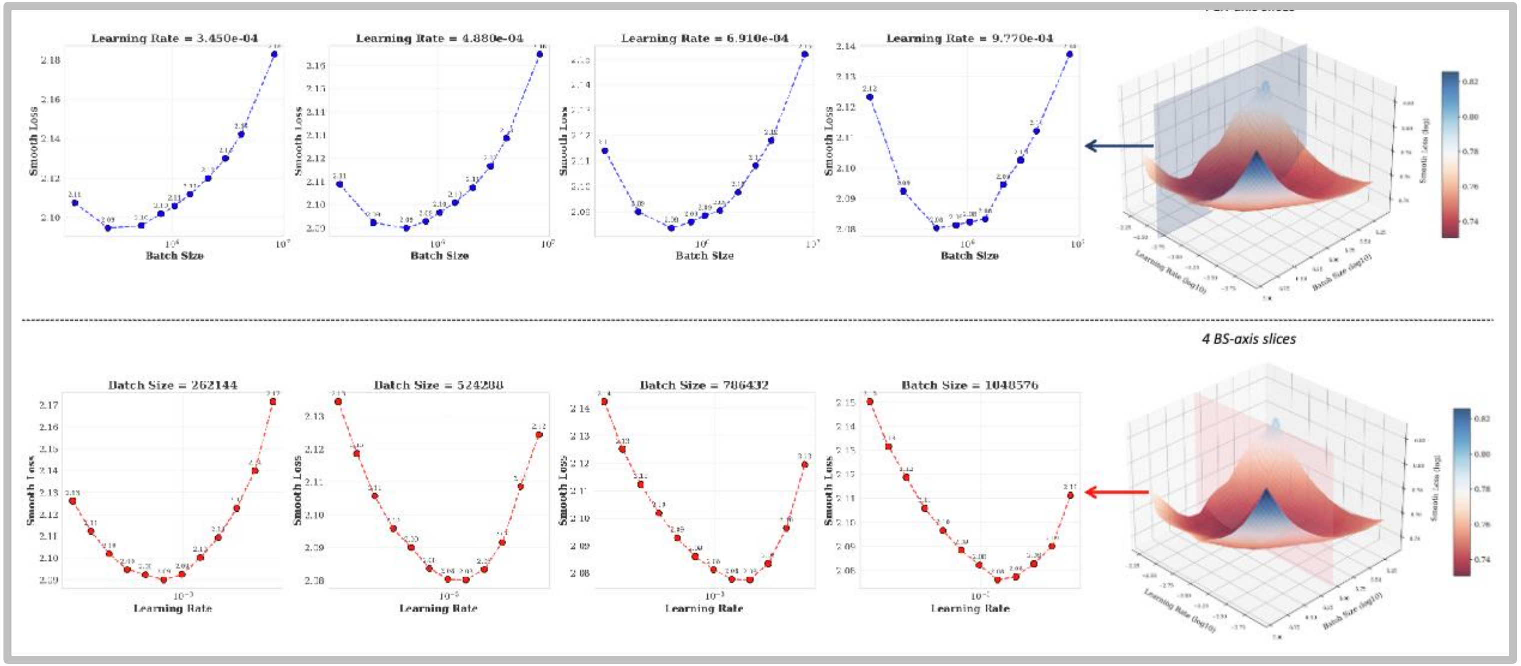
Much like the DeepSeek paper – train models to try to map out the hparam space



Model	N	D	d_{model}	d_{ff}	N_{head}	N_{layer}
1	2.15×10^8	1.14×10^{10}	960	9368	15	7
2	4.29×10^8	5.00×10^{10}	1280	9472	10	10
3	2.68×10^8	8.00×10^{10}	1024	9552	16	8
4	4.29×10^8	8.00×10^9	1280	9472	10	10
5	1.07×10^9	2.00×10^{10}	2048	8192	16	16
6	5.37×10^8	1.00×10^{10}	1280	9048	10	13
7	2.15×10^8	4.00×10^9	960	9368	15	7
8	2.68×10^8	5.00×10^9	1024	9552	16	8
9	2.68×10^8	1.42×10^{10}	1024	9552	16	8
10	1.07×10^9	5.69×10^{10}	2048	8192	16	16
11	2.15×10^8	1.00×10^{11}	960	9368	15	7
12	4.29×10^8	2.27×10^{10}	1280	9472	10	10
13	5.37×10^8	2.84×10^{10}	1280	9048	10	13
14	2.15×10^8	2.00×10^{10}	960	9368	15	7
15	4.29×10^8	4.00×10^{10}	1280	9472	10	10
16	2.68×10^8	2.50×10^{10}	1024	9552	16	8
17	5.37×10^8	5.00×10^{10}	1280	9048	10	13
18	1.07×10^9	1.00×10^{11}	2048	8192	16	16

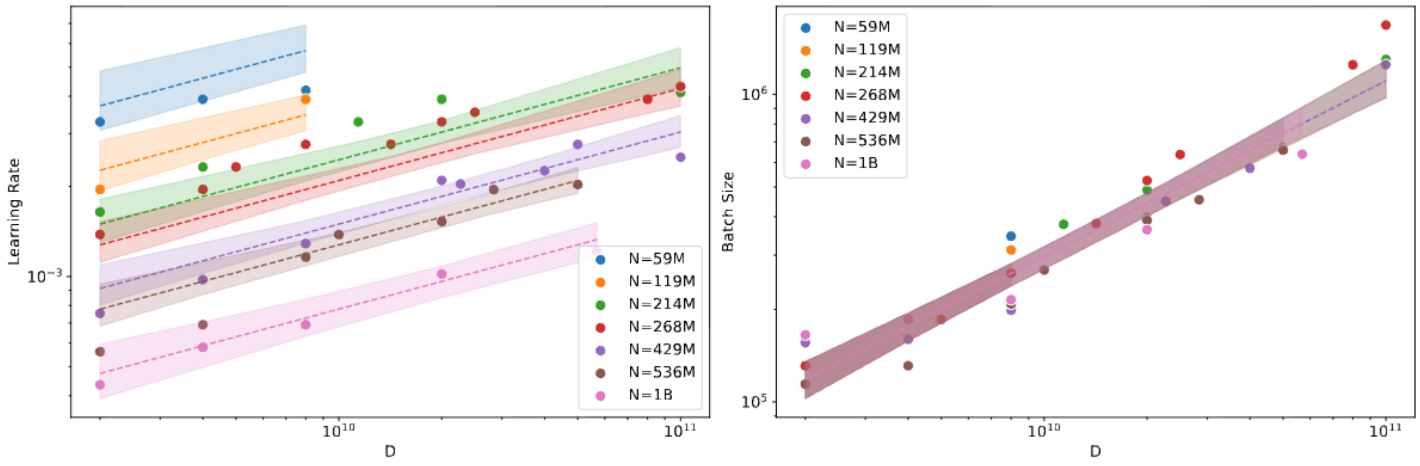
Table 5: **Dense Model Configuration.**

Observation 1: loss over batch/LR are convex



For pre-training losses, minimizers for LR/batch can be cleanly identified

Observation 2: scaling trends



(a) Scaling laws with dataset size for different number of parameters

For chinchilla style joint scaling: batch is primarily dependent on dataset size.

They also find higher optimal LR with D (for fixed M), but this is likely more fragile if swapping to WSD – see e.g. InternLM scaling law paper (Zhou+ 2026)

Observation 3 (?) robustness

Generalization to MoEs

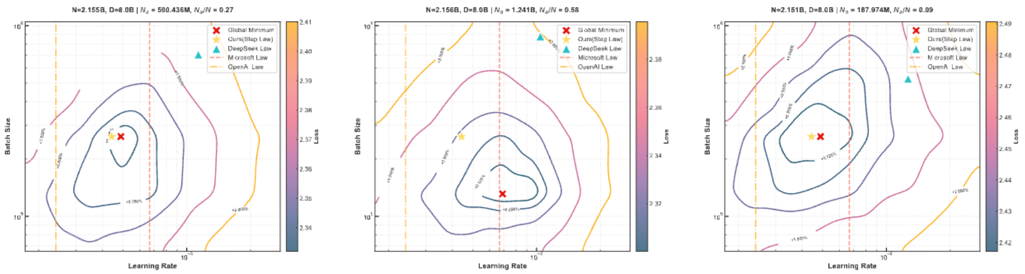
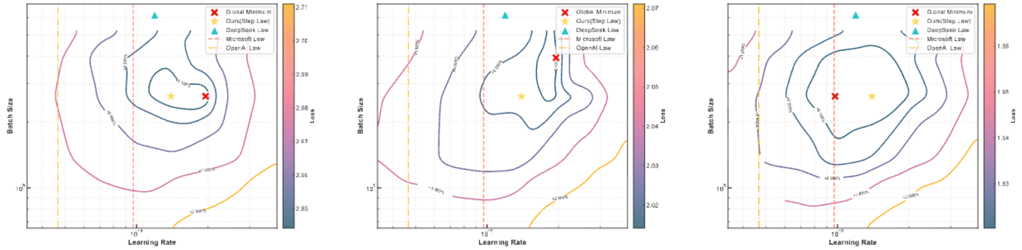


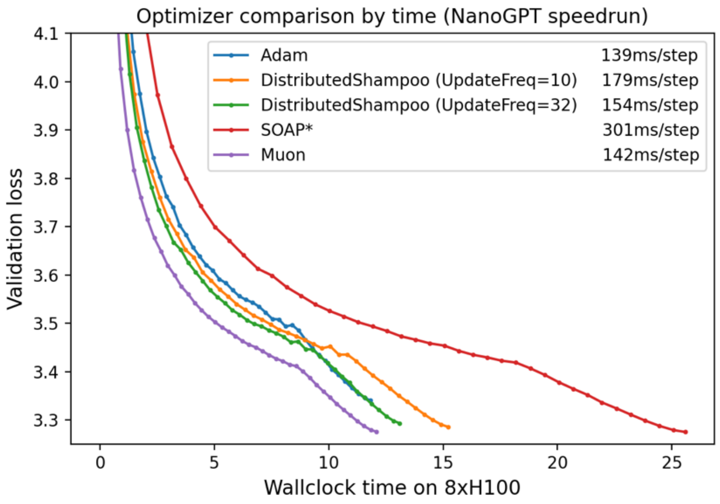
Figure 7: **Validation loss landscapes of MoE models under varying sparsity ratios** (N_a/N). Left: Low sparsity ($N_a/N = 0.27$). Middle: Medium sparsity ($N_a/N = 0.58$). Right: Medium sparsity at D=8.0B. Our method consistently approximates global minima across sparsity regimes.

Generalization to other datasets

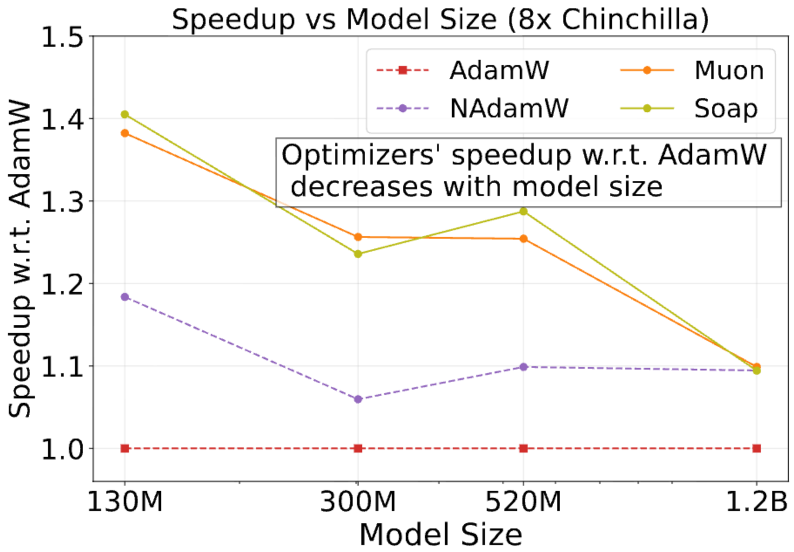


(a) Bilingual Corpus (b) Code Integration (c) Code-Dominante

Optimizers and scale



*SOAP is under active development. Future versions will significantly improve the wallclock overhead.



Optimization is a core part of LLMs (but also tricky, due to scale dependence!)

Fantastic Pretraining Optimizers and Where to Find Them

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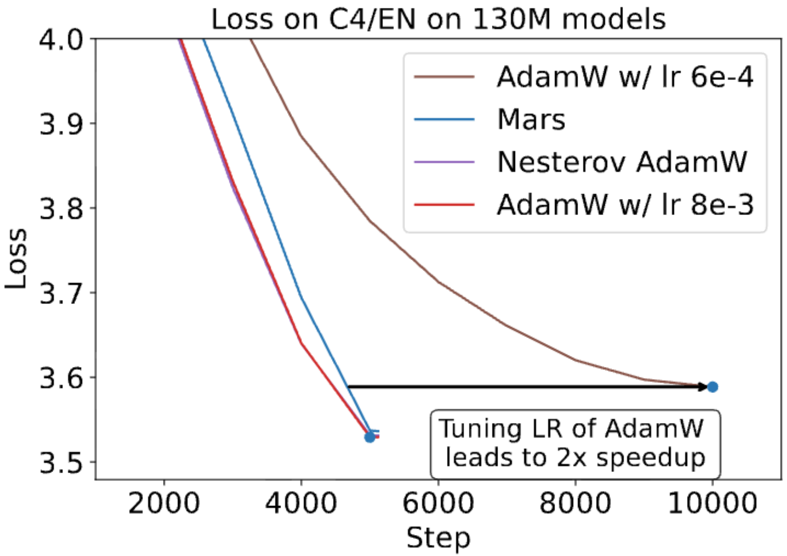
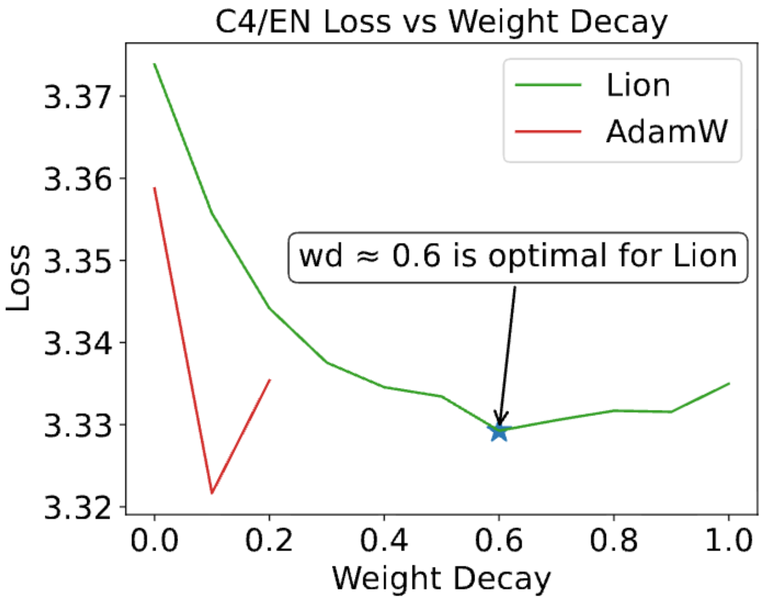
Stanford University

Problem 1 – hyper tuning is often off..

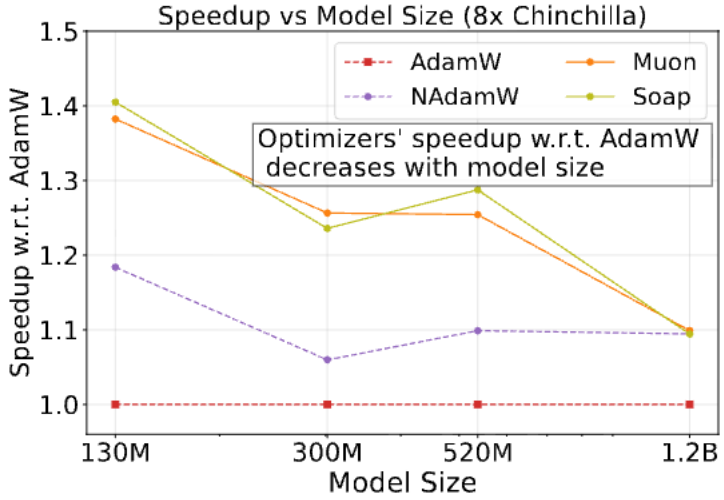


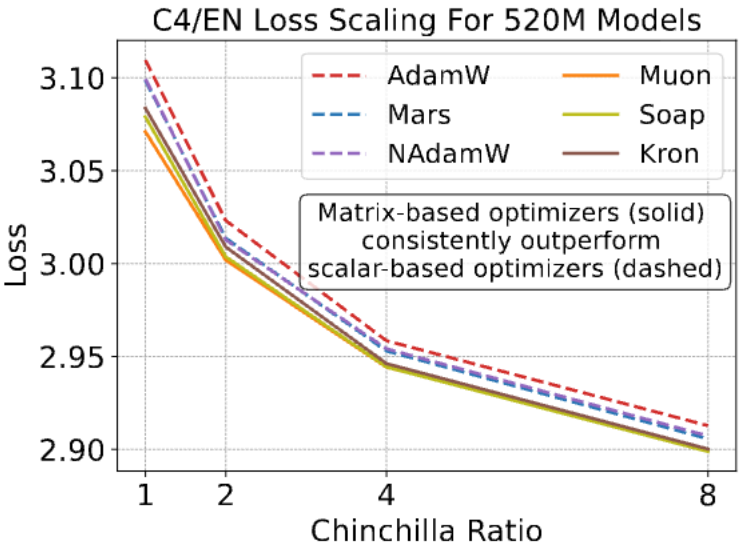


Different optimizers can require different hyperparameters!

(*and* likely have different optimal hyperparameter scaling)

Problem 2 – fairly significant scale dependence

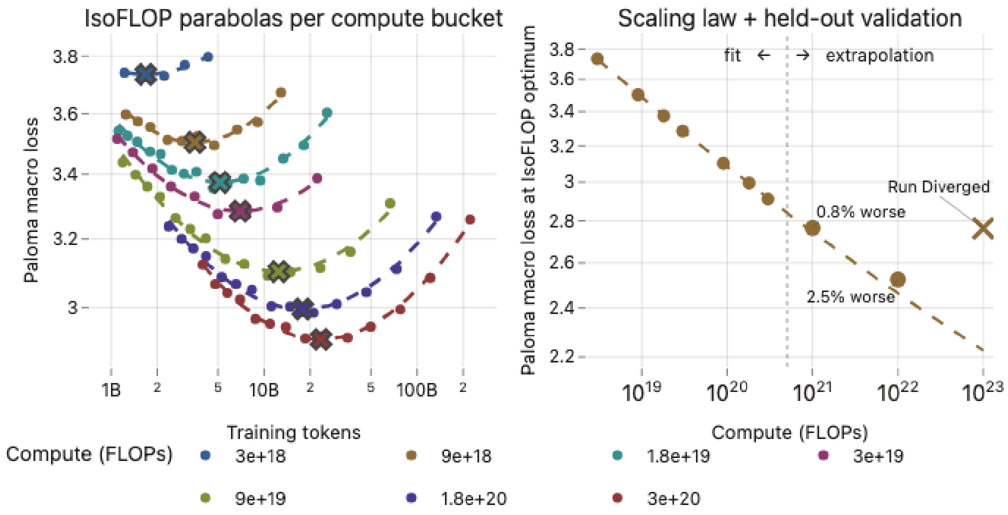




General algorithm development note – *always* check scaling with respect to compute and chinchilla ratios. These are often major confounders to performance!

Problem 2.5 – establishing scaling is nontrivial!

Sometimes, a good looking scaling can blow up



[From https://oa.williamheld.com/blog/delphi/]

Cautious AdamC + Sqrt batch-size scaling of learning rates
(Fixed with some more careful parametrization / scaling / optimizer changes)

A few slides on muon..

Muon:

Algorithm 2 Muon

Require: Learning rate η, momentum μ

Initialize $B_0 \leftarrow 0$

for $t = 1, \ldots$ **do**

  Compute gradient $G_t \leftarrow \nabla_\theta \mathcal{L}_t(\theta_{t-1})$

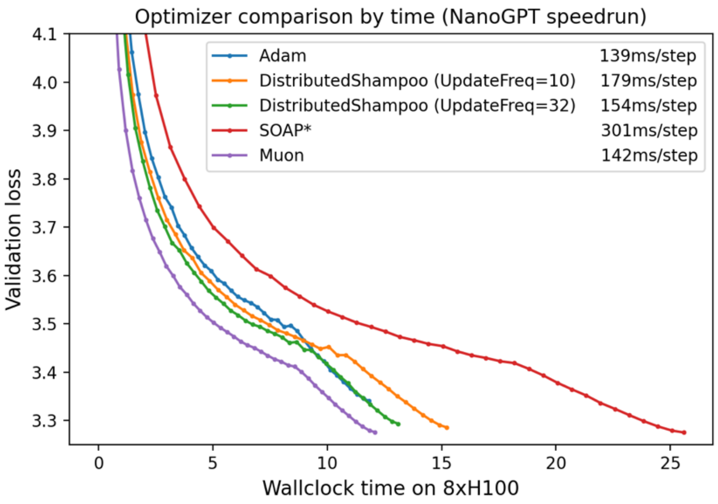
  $B_t \leftarrow \mu B_{t-1} + G_t$

  $O_t \leftarrow \text{NewtonSchulz5}(B_t)$

  Update parameters $\theta_t \leftarrow \theta_{t-1} - \eta O_t$

end for

return θ_t

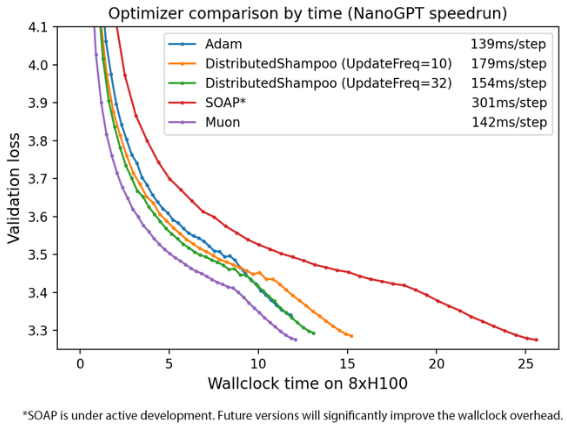


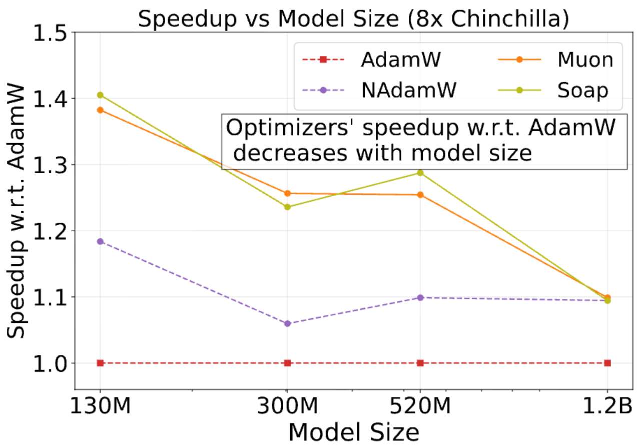
*SOAP is under active development. Future versions will significantly improve the wallclock overhead.

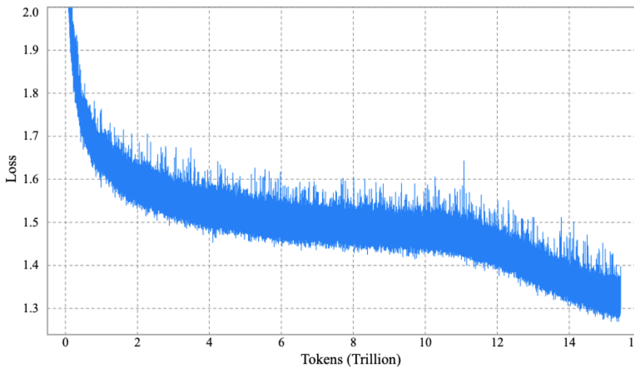
Optimizer for ‘matrix valued’ parameters.

NewtonSchultz (approximately) orthogonalizes the matrix $B_t = USV^\top \to UV^\top$

Muon and scaling







NanoGPT speedrun (very small!)

Scaling study

Kimi K2

Scaling gains are tricky to measure, but clearly muon ‘works’ at scale

Maximum update parametrization – in depth

Recall – the maximum update parametrization makes appealing claims

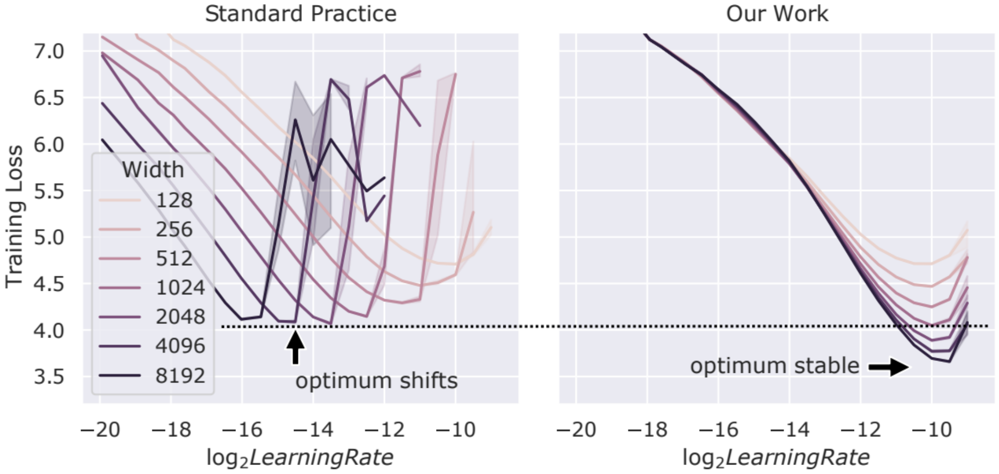


Table 2. μP function for a model M' that is r times the widths of M. If a parameter tensor has 2 dimensions that goes infinite when the model width goes infinite, it is "matrix-like" (*e.g.*, a fully-connected hidden layer); if the number is 1 or 0, it belongs to the "others" class. Note that embedding layers are "others". "Output" means the layer that maps an infinite dimension to a finite dimension, which is the word decoding layer (lm_head) in Transformers. A multiplier is a constant multiplied by a parameter tensor, which has a similar function to softmax temperature.

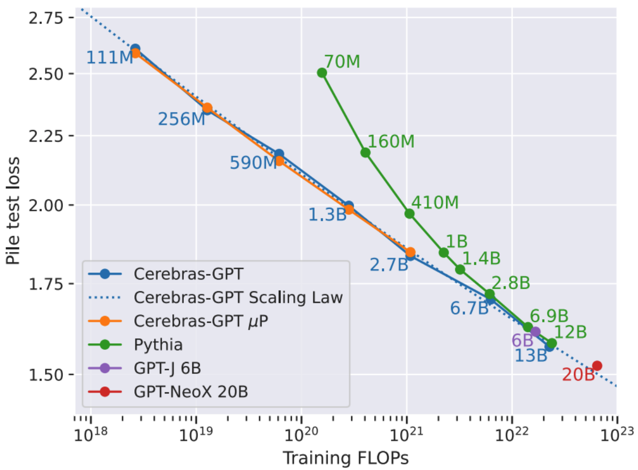
Hyperparameter (weight)	M	$M' \sim r$
AdamW learning rate (matrix-like)	l	l/r
AdamW learning rate (others)	l	l
Initialization variance (matrix-like)	σ	σ/r
Initialization variance (others)	σ	σ
Multiplier (output)	τ	τ/r
Multiplier (others)	τ	τ

Scale-invariant hyperparameter tuning would be very nice.

How does it work, and does it work in practice?

CerebrasGPT

CerebrasGPT – 0.1 to 13B models trained with the Chinchilla recipe.



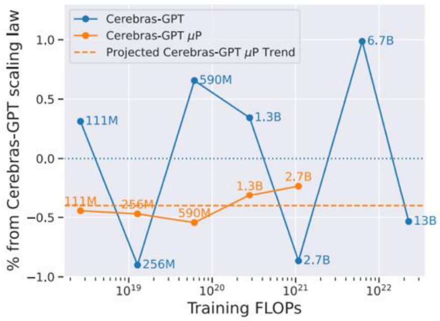


Figure 5: Percentage loss increase relative to Cerebras-GPT scaling law plotted against training FLOPs.

> We train a set of Cerebras-GPT models using µP. We follow the µTransfer approach by first tuning hyper-parameters for a small, 40M parameter µP model. Then, we transfer the hyperparameters along our µP scaling law up to a 2.7B parameter model. µP requires small changes to our baseline Cerebras-GPT models, including adding element-wise activation tensor scaling, adjusting initializers for affected layers, and adding layer-wise learning rates scaling to certain layers. We discuss the benefits we see with µP in Section 3.3. Refer to Appendix G for our tips to implement µP and our hyperparameter tuning notes.

Core finding – using muP parametrization makes scaling more stable

What is muP, anyway?

A Spectral Condition for Feature Learning

Greg Yang*
xAI

James B. Simon*
UC Berkeley & Imbue

Jeremy Bernstein*
MIT

(this is a very accessible 'muP for babies' paper)

muP is based off the following assertion. As a function of the width of the network n_l..

A1: The activations at initialization should remain $\Theta(1)$
A2: After one gradient step, the change in activation should be $\Theta(1)$

Note: if individual activations are $\Theta(1)$, then the norm should be $\Theta\left(\sqrt{n_l}\right)$

Deriving muP (condition A1)

Suppose that we have a simple, deep linear network ($h_l = W_l h_{l-1}$) and we init $W_l \sim N\left(0, \sigma^2 I_{n_l \times n_{l-1}}\right)$ then by basic matrix concentration $\|W_l\|_* \to \sigma(\sqrt{n_{l-1}} + \sqrt{n_l})$ and,

$$\|h_l\|_2 \approx \|W_l\|_* \|h_{l-1}\|_2$$

Now let's pick $\sigma = \frac{\sqrt{n_l}}{\sqrt{n_{l-1}}}\left(\sqrt{n_l} + \sqrt{n_{l-1}}\right)^{-1} = \Theta\left(\frac{1}{\sqrt{n_{l-1}}}\min\left(1, \sqrt{\frac{n_l}{n_{l-1}}}\right)\right)$. What happens?

Inductive assumption- $\|h_{l-1}\|_2 = \Theta\left(\sqrt{n_{l-1}}\right)$

Inductive case - $\|W_l\|_* \to \sigma\left(\sqrt{n_{l-1}} + \sqrt{n_l}\right) = \frac{\sqrt{n_l}}{\sqrt{n_{l-1}}}$

$$\|h_l\|_2 = \sqrt{n_l} + o(\sqrt{n_l})$$

[Comments – this is a kind of 'worst case' derivation – the ≈ is an upper bound]

Deriving muP (condition A2)

Now we need to deal with updates. Suppose we have the update ΔW_l on the weights. For SGD, on a linear layer, this looks like a rank-one loss-activation outer product.

$$\Delta W_l = -\eta_l \nabla_{h_l} \ell \; h_{l-1}^{\top}$$

Thus, $\|\Delta W_l h_{l-1}\|_2 = \|\Delta W_l\|_* \|h_{l-1}\|_2$. Now note that we have the update

$$\Delta \mathrm{h_l} = \mathrm{W_l} \Delta h_{l-1} + \Delta W_l (h_{l-1} + \Delta h_{l-1})$$

Assuming that the leading order terms don't cancel, we see that

- $\mathrm{W_l} \Delta h_{l-1} = \Theta(\sqrt{n_l})$ from induction assumption + condition A1 argument
- $\Delta W_l h_{l-1} = \|\Delta W_l\|_* \sqrt{n_{l-1}}$ from above, thus $\|\boldsymbol{\Delta W_l}\|_* = \boldsymbol{\Theta}\left(\frac{\sqrt{\boldsymbol{n_l}}}{\sqrt{\boldsymbol{n_{l-1}}}}\right)$
- $\Delta W_l \Delta h_{l-1} = O(\|\Delta W_l\|_* \sqrt{n_{l-1}})$

Deriving muP (condition A2) part 2

Recall –the key is to pick LR such that $\|\Delta W_l\|_* \sqrt{n_{l-1}} = \Theta(\sqrt{n_l})$. How can we do that?

Suppose that the loss update *also* scales O(1). Then we can write down..

$$\Delta \ell \approx \Theta(\langle \Delta W_l, \nabla_{W_l} \ell \rangle) = \Theta\left(\|\Delta W_l\|_F \left\|\nabla_{W_l} \ell\right\|_F\right) = \Theta\left(\|\Delta W_l\|_* \left\|\nabla_{W_l} \ell\right\|_*\right)$$

Where we use the fact that $\Delta W_l = -\eta \nabla_{W_l} \ell$ in standard SGD updates.

Now plug in $\Delta \ell = O(1)$, $\|\Delta W_l\|_* = \Theta\left(\frac{\sqrt{n_l}}{\sqrt{n_{l-1}}}\right)$ to get that $\left\|\nabla_{W_l} \ell\right\|_* = \Theta\left(\frac{\sqrt{n_{l-1}}}{\sqrt{n_l}}\right)$

Finally, from the previous slide, recall that $\Delta W_l = -\eta_l \nabla_{h_l} \ell \; h_{l-1}^\top$ and thus

$$\eta_l = \Theta\left(\frac{n_l}{n_{l-1}}\right)$$

[with Adam, $\|\Delta W_l\|_* \sqrt{n_{l-1}} = \Theta(\sqrt{\eta_l})$]

muP mini recap..

So, what is (baby) muP about? Controlling activations (and changes) via W and ΔW

Initialization (stdev): Set to $\Theta\left(\frac{1}{\sqrt{n_{l-1}}}\min\left(1, \sqrt{\frac{n_l}{n_{l-1}}}\right)\right)$

Learning rates: Set to $\frac{n_l}{n_{l-1}}$ (for Adam $\frac{1}{n_{l-1}}$)

Compared to 'standard' parametrizations – these set

Initialization: Set to $\frac{1}{\sqrt{n_{l-1}}}$

Learning rates: Set to $\Theta(1)$

Differences – LR changes for Adam, also init diffs when fanout n_l < fanin

Preprint

A Large-Scale Exploration of μ-Transfer

Lucas Dax Lingle
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Deeper dive into muP

Recall – muP is a scaling procedure for hyperparams (as a function of width)

Param	Init Variance (Θ)	Adam LR (Θ)	Init Variance (Exact)	Adam LR (Exact)
$\mathbf{W}^E$	1	1	1	α
$\mathbf{W}^{AQ}$	$1/M$	$1/M$	$1/M$	$\alpha P/M$
$\mathbf{W}^{AK}$	$1/M$	$1/M$	$1/M$	$\alpha P/M$
$\mathbf{W}^{AV}$	$1/M$	$1/M$	$1/M$	$\alpha P/M$
$\mathbf{W}^{AO}$	$1/(HD)$	$1/(HD)$	$1/M$	$\alpha P/M$
$\mathbf{W}^{FI}$	$1/M$	$1/M$	$1/M$	$\alpha P/M$
$\mathbf{W}^{FO}$	$1/F$	$1/F$	$0.25/M$	$\alpha P/M$
$\mathbf{W}^U$	$1/M^2$	$1/M$	$1/M^2$	$\alpha P/M$

Embedding

Attention params

Input/output MLP MM

Softmax linear

Table 2: μP scaling rules for transformers; a rule for attention scale is detailed in text below.

In addition, μP uses an attention scale of $\tau^{-1} = \Theta(1/D)$ instead of the usual $\tau^{-1} = 1/\sqrt{D}$. For simplicity, we use $\tau^{-1} = 1/D$, since in preliminary experiments we observed only a small improvement from using smaller multiples of $1/D$. Note that for D fixed across model widths M, any constant $\tau^{-1} \neq 0$ technically complies with μP (Yang et al., 2021) but in the experiments, τ^{-1} will be shown to have a major impact on performance and transfer.

Replicating muP

Q1: Does muP work as claimed? When we scale widths, is optimal LR constant?

Ablation	Width	Base LR					Transfer
		2^{-10}	2^{-8}	2^{-6}	2^{-4}	2^{-2}	
Baseline μP	128	3.846	3.743	**3.695**	3.884	4.143	✔
	512	3.114	2.993	**2.953**	3.221	3.506	
	2048	2.711	2.553	**2.511**	2.563	3.244	
Projection Biases	128	3.838	3.735	**3.705**	3.911	4.269	✔
	512	3.108	2.986	**2.947**	2.970	3.557	
	2048	2.710	2.552	**2.529**	2.672	3.418	

As shown in Table 1, the learning rates transfer reliably across model sizes under μP. Despite each model being 4x wider (and 16x larger) than the last, the smallest model's optimal base learning rate α directly predicts the optimum in our sweeps for the larger models.

What is muP robust to?

Modern LMs have many components that deviate from muP's theory

- Activations – SwiGLU and squared relu
- Batch sizes – Large / small
- Initialization variations – zero attention, etc.
- RMS norm gains
- Exotic optimizers (Lion)
- Regularizers

Which of these (if any) break muP?

What is muP not robust to? RMSnorm gain

In our arch – RMSNorm has learnable gains. This turns out to break muP

Ablation	Width	Base LR					Transfer
		2^{-10}	2^{-8}	2^{-6}	2^{-4}	2^{-2}	
Baseline μP	128	3.846	3.743	**3.695**	3.884	4.143	✔
	512	3.114	2.993	**2.953**	3.221	3.506	
	2048	2.711	2.553	**2.511**	2.563	3.244	
RMSNorm Gains (Vector)	128	3.842	3.744	3.689	**3.670**	3.681	✗
	512	3.101	2.992	2.951	**2.950**	3.412	
	2048	2.692	**2.553**	2.609	2.605	3.169	
RMSNorm Gains (Scalar)	128	3.843	3.749	3.692	**3.670**	4.471	✗
	512	3.106	3.000	2.961	**2.959**	3.515	
	2048	2.704	2.570	**2.525**	2.542	3.334	

As shown in Table 1, optimal learning rates for these models *do not* reliably transfer when using $\Theta(1)$ learning rate scaling for the gains, despite the fact that the 'coordinate size' of the features before and after RMS Normalization is $\Theta(1)$ w.r.t. width by design. In addition to the lack of transfer in these experiments, we find trainable gains harm the quality of the largest μP models when the base learning rate α is optimal. In Section 4.5, we also find that standard transformers with trainable gains underperform μP transformers without them.

But these gains can be removed with little loss of perf..

What is muP not robust to? Exotic optimizers

There are other, exotic optimizers based on just gradient signs. Do they transfer?

Algorithm 1 AdamW Optimizer

given $\beta_1, \beta_2, \epsilon, \lambda, \eta, f$
initialize $\theta_0, m_0 \leftarrow 0, v_0 \leftarrow 0, t \leftarrow 0$
while θ_t not converged **do**
  $t \leftarrow t + 1$
  $g_t \leftarrow \nabla_\theta f(\theta_{t-1})$
  update EMA of g_t **and** g_t^2
  $m_t \leftarrow \beta_1 m_{t-1} + (1 - \beta_1) g_t$
  $v_t \leftarrow \beta_2 v_{t-1} + (1 - \beta_2) g_t^2$
  bias correction
  $\hat{m}_t \leftarrow m_t / (1 - \beta_1^t)$
  $\hat{v}_t \leftarrow v_t / (1 - \beta_2^t)$
  update model parameters
  $\theta_t \leftarrow \theta_{t-1} - \eta_t(\hat{m}_t / (\sqrt{\hat{v}_t} + \epsilon) + \lambda\theta_{t-1})$
end while
return θ_t

Algorithm 2 Lion Optimizer (ours)

given $\beta_1, \beta_2, \lambda, \eta, f$
initialize $\theta_0, m_0 \leftarrow 0$
while θ_t not converged **do**
  $g_t \leftarrow \nabla_\theta f(\theta_{t-1})$
  update model parameters
  $c_t \leftarrow \beta_1 m_{t-1} + (1 - \beta_1) g_t$
  $\theta_t \leftarrow \theta_{t-1} - \eta_t(\text{sign}(c_t) + \lambda\theta_{t-1})$
  update EMA of g_t
  $m_t \leftarrow \beta_2 m_{t-1} + (1 - \beta_2) g_t$
end while
return θ_t

Lion Optimizer	128	**3.708**	3.736	4.057	4.344	10.380	✗
	512	2.952	**2.947**	3.416	3.961	10.285	
	2048	2.519	**2.511**	3.151	10.377	10.377	

What is muP not robust to? – (strong) weight decay

What about strong (0.1) weight decay? – this is maybe the only significant muP failure

Decoupled Weight Decay	128	3.760	**3.679**	3.694	3.741	4.011	✗
	512	3.057	2.963	**2.957**	3.139	3.373	
	2048	2.686	2.535	**2.502**	3.123	6.594	

Algorithm 1 SGD with L_2 regularization and SGD with decoupled weight decay (SGDW), both with momentum

1: **given** initial learning rate $\alpha \in \mathbb{R}$, momentum factor $\beta_1 \in \mathbb{R}$, weight decay/L_2 regularization factor $\lambda \in \mathbb{R}$
2: **initialize** time step $t \leftarrow 0$, parameter vector $\boldsymbol{\theta}_{t=0} \in \mathbb{R}^n$, first moment vector $\boldsymbol{m}_{t=0} \leftarrow \boldsymbol{0}$, schedule multiplier $\eta_{t=0} \in \mathbb{R}$
3: **repeat**
4: $\quad t \leftarrow t + 1$
5: $\quad \nabla f_t(\boldsymbol{\theta}_{t-1}) \leftarrow \text{SelectBatch}(\boldsymbol{\theta}_{t-1})$ ▷ select batch and return the corresponding gradient
6: $\quad \boldsymbol{g}_t \leftarrow \nabla f_t(\boldsymbol{\theta}_{t-1}) \; +\lambda\boldsymbol{\theta}_{t-1}$
7: $\quad \eta_t \leftarrow \text{SetScheduleMultiplier}(t)$ ▷ can be fixed, decay, be used for warm restarts
8: $\quad \boldsymbol{m}_t \leftarrow \beta_1 \boldsymbol{m}_{t-1} + \eta_t \alpha \boldsymbol{g}_t$
9: $\quad \boldsymbol{\theta}_t \leftarrow \boldsymbol{\theta}_{t-1} - \boldsymbol{m}_t \; -\eta_t\lambda\boldsymbol{\theta}_{t-1}$
10: **until** *stopping criterion is met*
11: **return** optimized parameters $\boldsymbol{\theta}_t$

Is muP useful? At least to some extent..

Overall, muP generally seems useful – insofar that SP is quite a bit more unstable.

Width	LR				
	2^{-10}	2^{-8}	2^{-6}	2^{-4}	2^{-2}
128	3.841	3.757	**3.706**	3.879	4.030
512	3.013	**2.967**	2.987	3.383	7.403
2048	**2.738**	2.902	7.247	7.477	7.314

Table 3: Validation losses for SP models.

Params	Width	Base LR		
		2^{-8}	2^{-6}	2^{-4}
2M	128	3.791	**3.766**	3.814
40M	512	3.016	**2.983**	3.004
600M	2048	2.513	**2.459**	2.466
10B	8192	2.238	**2.167**	2.169

Table 4: Validation losses for our large-scale experiment.

Current evidence suggests that muP parametrization / initialization may be easier to tune.

Recap: scaling in the wild

What are challenges in scaling ‘in practice’

1. Setting model arch hyperparameters (width, etc)
2. Setting optimizer hyperparameters (LR, batch)
3. Compute needed to fit the big chinchilla sweep

Some solutions?

1. Assume stability (or use muP)
2. Search for optimal LR / batch in small scale, either keep fixed or predict scaling
3. Use alternative learning schedules (WSD-like)